Calculus of Functions of One and Several Variables-01/02/03 (FA24)

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Math 008: Calculus of Functions of One and Several Variables

Instructors

Instructor	Juliette Bruce	Friedrich Bauermeister
Section	01	03
Class	(10) MWF: 10:10am - 11:15pm	(2) MWF: 2:10pm - 3:15pm
X-Hour	Th: 12:15pm - 1:05pm	Th: 1:20am - 2:10am
Classroom	Kemeny 007	Kemeny 007
Contact	juliette.bruce@dartmouth.edu (mailto:juliette.bruce@dartmouth.edu)	friedrich.bauermeister.gr@dartmouth.edu (mailto:%20friedrich.bauermeister.gr@dartmouth.edu
Office	Kemeny 342	
Office Hours	Th 12:15pm - 1:05pm (Office), W: 2:30pm - 3:30pm (Office)	

Course Meetings:

- **In-Person Lectures:** All lectures will be held in person unless otherwise stated. They will not be recorded, nor will slides or lecture notes be available. If you are not feeling well or have been instructed not to come to class, please get in touch with the instructor prior to class. In this case, the instructor will try to arrange to have a classmate take notes for you. Lectures may be rescheduled to the X-hour if needed.
- **In-Person & Remote Office Hours:** Office hours will generally be held in-person. From time to time, it may be announced in class or on CANVAS that office hours will be conducted via Zoom. Individual appointments with instructors may be held remotely via Zoom.

You are currently logged into Student View

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course as a brand new

Thursdays from 7:00-9:00pm in Kemeny 007, focusing on answering your questions on the homework and class material. For maximal benefit, we strongly recommend that you try all the homework problems ahead of time and come with your questions to the tutorial. Tutorials are open to all Math 8 students. You don't need an appointment.

- **X-Hours:** The X-hours will be used and scheduled as needed to supplement, extend, or make-up our normal course lectures at my discretion. When scheduled students are expected to attended the X-hours.

Textbook:

- *Calculus (8th Edition)* By James Stewart

Prerequisites:

- Math 3 or advanced placement into Math 8.

Grading:

- **Grade Breakdown:** Grades will be based on:
  - Weekly Homework (20%),
  - Class Attendance (10%),
  - Midterm Exams (40%), and
  - Final Exam (30%)

Midterm and Final Exams:

- **Midterm Exam #1:** Monday, October 7, 2024 at 5:00pm - 7:00pm; Location Wilder 104
- **Midterm Exam #2:** Monday, October 28, 2024 at 5:00pm - 7:00pm; Location Wilder 104
- **Final Exam:** Friday, November 22, 2024 at 3:00pm-5:00pm; Location Wilder 104. (Note that we cannot accommodate travel plans for the final exam and you are expected to be here on this date.)

Homework:

We will be using webwork (through canvas). We will also suggest textbook problems which will be

posted **here (https://canvas.dartmouth.edu/courses/67810/pages/suggested-problems)** . Homework reinforces concepts and skills while challenging students to extend what they have learned to other types of problems. Because it is important for students to have this experience, instructors will not solve assigned homework problems during office hours before the due date. We will of course answer questions you may have in approaching problems that give you difficulty. It is therefore essential to begin homework sets early so that you may get help if difficulties do arise.

Homework is due each Tuesday at 11:59 pm. We are using **webwork** , which will be administered through your CANVAS website. As all homework is posted well in advance, no late homework will be accepted. Homework typically covers course material through the past week.

The lowest homework score will be dropped.

Attendance:

Attendance is a crucial aspect of this course, and students are expected to attend, be engaged with, and contribute class. As an indicator of the importance of being engaged and reflective in the course, attendance counts for 10% of the overall grade. Attendance is measure by completing weekly reflection assignments that ask you write one paragraph each week reflecting on the course. These assignments are due on Fridays at 11:59pm and are submitted via Canvas. No late reflections will be accepted.

The lowest reflection score will be dropped.

Tentative Lecture Plan:

The course schedule below is tentative and will be updated as we proceed through the semester:

Class Meeting	Class Day	Topics	Textbook Sections
Week 1.1	9/16 (M)	Differentiation/Integration Review, Infinite Sequences/Series	11.1, 11.2
Week 1.2	9/18 (W)	Infinite Series, Integral Test	11.2,11.3
Week 1.3	9/20 (F)	Integral Test, Comparison Tests	11.3, 11.4

Week 2.1	9/23 (M)	Alternating Series, Ratio Test	11.5, 11.6
Week 2.2	9/25 (W)	Power Series, Radius of Convergence	11.8
Week 2.3	9/27 (F)	Functions as power series	11.9
Week 3.1	9/30 (M)	Taylor Series	11.10
Week 3.2	10/2 (W)	Taylor Series	11.10
Week 3.3	10/4 (F)	Midterm #1 Review	
Week 4.1	10/7 (M)	Midterm Exam #1 (5:00pm - 7:00pm)	
Week 4.1	10/7 (M)	Coordinates and Vectors	12.1, 12.2
Week 4.2	10/9 (W)	Dot Product & Projection	12.3
Week 4.3	10/11 (F)	Dot Product & Projection	12.3
Week 5.1	10/14 (M)	Cross Product & Geometry	12.4
Week 5.2	10/16 (W)	Representing Lines & Planes	12.5
Week 5.3	10/18 (F)	Representing Lines & Planes	12.5
Week 6.1	10/21 (M)	Vector Functions	13.1
Week 6.2	10/23 (W)	Derivatives and Integrals of Vector Functions	13.2
Week 6.3	10/25 (F)	Midterm #2 Review	
Week 7.1	10/28 (M)	Midterm Exam #2 (5:00pm - 7:00pm)	
Week 7.1	10/28 (M)	Functions of Several Variables	14.1, 14.2
Week 7.2	10/30 (W)	Partial Derivatives	14.3
Week 7.3	11/1 (F)	Tangent Planes	14.4
Week 8.1	11/4 (M)	Chain Rule	14.5

Week 8.2	11/6 (W)	Gradient & Directional Derivatives	14.6
Week 8.3	11/8 (F)	Gradient & Directional Derivatives	14.6
Week 9.1	11/11 (M)	Mins & Maxs	14.7
Week 9.2	11/13 (W)	Lagrange Multipliers	14.8
Week 9.3	11/15 (F)	Final Review	
Week 10.1	11/22 (F)	Final Exam (3:00pm - 5:00pm)	

Course Expectations:

I believe in the axioms laid out by SFSU Professor* ***Federico Ardila*** ***(http://fardila.com/)*** *, and I will use them to guide my instruction of this course.*

1. Mathematical potential is distributed equally among different groups, irrespective of geographic, demographic, and economic boundaries.

2. Everyone can have joyful, meaningful, and empowering mathematical experiences.

3. Mathematics is a powerful, malleable tool that can be shaped and used differently by various communities to serve their needs.

4. Every student deserves to be treated with dignity and respect.

*Mathematically, an axiom is a statement accepted as truth, on which all other work is built.

Using these as guiding principles, I expect students in Math 8 to collaborate with me in developing and maintaining an inclusive learning environment where diversity and individual differences are understood, respected, and recognized as a source of strength. Racism, discrimination, harassment, and bullying will not be tolerated. I expect all participants in this course (students and faculty alike) to treat each other with courtesy and respect.

Mathematics requires active participation. Before each class period, please read the assigned sections. Arrive to class ready to share what you have learned and also what remains confusing. Class meetings will involve some interactive lecture and other activities in a variety of formats; you will get the most out of each class day if you arrive ready to engage. In all settings, collaborate thoughtfully and ask questions respectfully: everyone should be able to freely participate.

Honor Principle

We will strictly enforce Dartmouth's **Academic Honor Principle (https://students.dartmouth.edu/community-standards/policy/academic-honor-principle/)** . Please be advised of especially

- **Exams**: Giving and/or receiving assistance during an examination violates the Academic Honor Principle.
- **Homework**: Collaboration is both permitted and encouraged, but it is a violation of the honor code for someone to provide the answers for you.

Course Catalogue Description:

This course is a sequel to MATH 3 and is appropriate for students who have successfully completed an AB calculus curriculum (or the equivalent) in secondary school. Roughly half of the course is devoted to topics in one-variable calculus, selected from techniques of integrations, areas, volumes, numerical integration, sequences and series including Taylor series, ordinary differential equations and techniques of their solution. The second half of the course studies scalar valued functions of several variables. It begins with the study of vector geometry, equations of lines and planes, and space curves (velocity, acceleration, arclength). The balance of the course is devoted to studying differential calculus of functions of several variables. Topics include limits and continuity, partial derivatives, tangent planes and differentials, the Chain Rule, directional derivatives and applications, and optimization problems including the use of Lagrange multipliers.

Mental Health:

The academic environment at Dartmouth is challenging, our terms are intensive, and classes are not the only demanding part of your life. There are a number of resources available to you on campus to support your wellness:

- Undergraduate Dean (**http://www.dartmouth.edu/~upperde/ (http://www.dartmouth.edu/~upperde/)**);
- Counseling and Human Development (**http://www.dartmouth.edu/~chd/ (http://www.dartmouth.edu/~chd/)**); and the
- Student Wellness Center (**http://www.dartmouth.edu/~healthed/ (http://www.dartmouth.edu/~healthed/)**).

I encourage you to use these resources to take care of yourself throughout the term. Please come

speak to me if you experience any difficulties, or would like help accessing any of these recourses.

Student Accessibility Services:

Students requesting disability-related accommodations and services for this course are encouraged to schedule a phone/video meeting with me as early in the term as possible. This conversation will help to establish what supports are built into my online course. In order for accommodations to be authorized, students are required to consult with **Student Accessibility Services (https://students.dartmouth.edu/student-accessibility/)** (SAS; **student.accessibility.services@dartmouth.edu (mailto:student.accessibility.services@dartmouth.edu)** ; 603-646-9900) and to email me their SAS accommodation form. We will then work together with SAS if accommodations need to be modified based on the online learning environment. If students have questions about whether they are eligible for accommodations, they should contact the SAS office. All inquiries and discussions will remain confidential.

Title IX:

At Dartmouth, we value integrity, responsibility, and respect for the rights and interests of others, all central to our Principles of Community. We are dedicated to establishing and maintaining a safe and inclusive campus where all have equal access to the educational and employment opportunities Dartmouth offers. We strive to promote an environment of sexual respect, safety, and well-being. In its policies and standards, Dartmouth demonstrates unequivocally that sexual assault, gender-based harassment, domestic violence, dating violence, and stalking are not tolerated in our community.

The Sexual Respect Website (**https://sexual-respect.dartmouth.edu (https://sexual-respect.dartmouth.edu/)**) at Dartmouth provides a wealth of information on your rights with regard to sexual respect and resources that are available to all in our community.

Please note that, as a faculty member, I am obligated to share disclosures regarding conduct under Title IX with Dartmouth's Title IX Coordinator. Confidential resources are also available, and include licensed medical or counseling professionals (e.g., a licensed psychologist), staff members of organizations recognized as rape crisis centers under state law (such as WISE), and ordained clergy (see **https://dartgo.org/titleix_resources (https://dartgo.org/titleix_resources)**).

Should you have any questions, please feel free to contact Dartmouth's Title IX Coordinator or the Deputy Title IX Coordinator for the Guarini School. Their contact information can be found on the sexual respect website at: **https://sexual-respect.dartmouth.edu (https://sexual-respect.dartmouth.edu/)** .

Religous Observances:

Some students may wish to take part in religious observances that occur during this academic term. If you have a religious observance that conflicts with your participation in the course, please meet with me as soon as possible, or before the end of the second week of the term (at the latest) to discuss appropriate adjustments. Dartmouth has a deep commitment to support students' religious observances and diverse faith practices.

Additional Support:

Cooperation on weekly homework is permitted (and encouraged), but you should write up the solution on your own. In other words, share your thoughts (say on a blackboard) to get started, but write up the solution afterwards using only your understanding. Please acknowledge any cooperative work at the end of each assignment.

Plagiarism, collusion, or other violations of the **Academic Honor Principle (https://students.dartmouth.edu/community-standards/policy/academic-honor-principle)** will be referred to the The Committee on Standards. If you have any questions as to whether some action would be acceptable under the Academic Honor Principle, please speak to me beforehand. For more information about standards of conduct, please see **these resources (https://students.dartmouth.edu/community-standards/)** .

Course Summary:

Date	Details	Due
Fri Sep 20, 2024	**Week 1 Reflection (https://canvas.dartmouth.edu/courses/67810/assignments/481302)**	due by 11:59pm
Tue Sep 24, 2024	**Homework #1 (https://canvas.dartmouth.edu/courses/67810/assignments/481642)**	due by 11:59pm

Fri Sep 27, 2024	Week 2 Reflection (https://canvas.dartmouth.edu/courses/67810/assignments/481303)	due by 11:59pm
Tue Oct 1, 2024	Homework #2 (https://canvas.dartmouth.edu/courses/67810/assignments/481643)	due by 11:59pm
Fri Oct 4, 2024	Week 3 Reflection (https://canvas.dartmouth.edu/courses/67810/assignments/481310)	due by 11:59pm
Mon Oct 7, 2024	Midterm Exam #1 (https://canvas.dartmouth.edu/courses/67810/assignments/480995)	due by 5pm
Tue Oct 8, 2024	Homework #3 (https://canvas.dartmouth.edu/courses/67810/assignments/481644)	due by 11:59pm
Fri Oct 11, 2024	Week 4 Reflection (https://canvas.dartmouth.edu/courses/67810/assignments/481315)	due by 11:59pm
Tue Oct 15, 2024	Homework #4 (https://canvas.dartmouth.edu/courses/67810/assignments/486398)	due by 11:59pm
Fri Oct 18, 2024	Week 5 Reflection (https://canvas.dartmouth.edu/courses/67810/assignments/481317)	due by 11:59pm
Tue Oct 22, 2024	Homework #5 (https://canvas.dartmouth.edu/courses/67810/assignments/486400)	due by 11:59pm
Fri Oct 25, 2024	Week 6 Reflection (https://canvas.dartmouth.edu/courses/67810/assignments/481318)	due by 11:59pm
Mon Oct 28, 2024	Midterm Exam #2 (https://canvas.dartmouth.edu/courses/67810/assignments/480996)	due by 5pm
Tue Oct 29, 2024	Homework #6 (https://canvas.dartmouth.edu/courses/67810/assignments/486401)	due by 11:59pm
Fri Nov 1, 2024	Week 7 Reflection (https://canvas.dartmouth.edu/courses/67810/assignments/481319)	due by 11:59pm

Tue Nov 5, 2024	**Homework #7** (https://canvas.dartmouth.edu/courses/67810/assignments/489177)	due by 11:59pm
Fri Nov 8, 2024	**Week 8 Reflection** (https://canvas.dartmouth.edu/courses/67810/assignments/481320)	due by 11:59pm
Tue Nov 12, 2024	**Homework #8** (https://canvas.dartmouth.edu/courses/67810/assignments/489179)	due by 11:59pm
Fri Nov 15, 2024	**Week 9 Reflection** (https://canvas.dartmouth.edu/courses/67810/assignments/481321)	due by 11:59pm
Mon Nov 18, 2024	**Homework #9** (https://canvas.dartmouth.edu/courses/67810/assignments/489180)	due by 11:59pm
Mon Nov 25, 2024	**Final Exam** (https://canvas.dartmouth.edu/courses/67810/assignments/480997)	due by 8am